Nahian Tahmin, Ph.D.

Data Scientist | Healthcare and Life Sciences Analytics

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PROFESSIONAL SUMMARY

Data scientist with a Ph.D. in Data Science and Engineering and peer-reviewed work in healthcare predictive modeling. Turns high-dimensional, multi-source data into validated machine learning models and explains what drives them, using Python, SQL, R, PyTorch, and scikit-learn. Experienced in statistical analysis, feature engineering, data cleaning, and data visualization, and in communicating results to clinicians, geneticists, engineers, and other stakeholders. Methods transfer to other domains with imbalanced or multi-source data.

EDUCATION

University of Tennessee, Knoxville Aug 2022 - Aug 2026

Ph.D., Data Science and Engineering. GPA 3.95/4.00.

Monash University Malaysia Oct 2017 - Oct 2021

B.Eng., Software Engineering.

PUBLICATIONS

Tahmin N, Chinthala L, Marsico F, Buonaiuto S, Mohammed A, Carlisle A, Gautam Y, Colonna V, Mersha T, Davis RL, Khojandi A. Machine learning models incorporating genotype and ancestry improve severe asthma risk prediction. *Scientific Reports* 15, 40243 (2025). doi.org/10.1038/s41598-025-24080-x

Tahmin N, Chinthala LK, Mersha TB, Davis RL, Khojandi A. X-Admix: An Interpretable Multimodal Cross-Attention Framework for Integrating Genotype, Local Ancestry, and Social Drivers of Health in Admixed African American Populations. *medRxiv* preprint (2026); under review at PLOS Computational Biology. doi.org/10.64898/2026.09.03.26362179

Pokhrel D, **Tahmin N**, Carlisle A, Mersha T, Davis RL, Khojandi A. Assessing the Role of Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis. *IEEE SIEDS* (2025). doi.org/10.1109/SIEDS65500.2025.11021139

Predicting Precision Inhaled Corticosteroids Response in Asthma Patients. Presented at the ASHG Annual Meeting 2023, Abstract 2023-A-1763-ASHG.

Integrating genomics, local ancestry, and social drivers of health (SDOH) on asthma medication response prediction. Presented at ASHG Annual Meeting 2023 and INFORMS Annual Meeting 2024 and 2025.

PROJECTS

Predictive Modeling on Multi-Source Healthcare and Genomic Data Aug 2022 - Jul 2026

University of Tennessee, Knoxville

- **Risk prediction from multi-source data.** Built a deep learning model that combines genetic, ancestry, and social drivers of health (SDOH) data. Reached AUC 0.771 on a clinical outcome prediction task (asthma) where standard machine learning models stayed near chance (AUC 0.566 or lower), with the trend confirmed on an independent All of Us cohort roughly 3x larger.
- **Treatment response modeling.** Predicted inhaled corticosteroid response in a pediatric cohort (N = 248, about 5 million candidate features). Used Lasso, RFE, and Elastic Net with expert priors for feature selection; the stacked model improved on single-source baselines by 5 AUC points. Published in Scientific Reports (2025).
- **Explaining model drivers.** Combined feature ablation, gradient-based attribution, and SHAP, and ranked findings by reproducibility across cross-validation folds so that only stable drivers were reported.
- **LLMs on structured data (completed as part of doctoral dissertation, 2026).** Tested four ways of encoding tabular records for pretrained large language models under a fixed token budget; compressed encodings beat verbose text (AUC 0.71 vs. 0.63).

Social Determinants in Asthma Across Racial Groups: A Comparative Model Analysis

- Mentored a junior Ph.D. student to build interpretable models integrating demographic, socioeconomic, and clinical features, using targeted ablations to identify subgroup-specific SDOH drivers and translate findings into insights. Oversaw end-to-end study design; published at IEEE SIEDS 2025.

Advisor, Vertically Integrated Program (VIP) Team, Criminal Sentences Project, *Human-Augmented Analytics Group*

- Advised a cross-functional student team building Named Entity Recognition (NER) models that extract sentencing components from federal court records; supervised evaluation criteria and error analysis.

SKILLS

Modeling: Predictive modeling, machine learning, deep learning, statistical techniques, feature selection (Lasso, Elastic Net, RFE), Random Forest, ensemble stacking, Transformers, LLM fine-tuning

Validation and Explainability: Cross-validation, hyperparameter tuning, ablation analysis, SHAP, Captum, subgroup analysis
Programming and Data: Python (Pandas, NumPy, scikit-learn, PyTorch, TensorFlow), SQL, R, PySpark, data analysis, data cleaning, feature engineering
Tools: Power BI, Google Cloud, Oracle, Git/GitHub, Weights and Biases, Hugging Face

PROFESSIONAL EXPERIENCE

University of Tennessee, Knoxville, *Graduate Research Assistant* Aug 2022 - Jul 2026
Designed scalable, reliable, and interpretable AI, deep learning, and LLM research workflows; data engineering and fusion for multimodal datasets; worked in a cross-disciplinary team.

ITQAN Analytics and Software Limited, *Developer Intern* Mar 2022 - Jul 2022
Developed Python data engineering pipelines and Power BI dashboards for data visualization.

Monstar Lab Bangladesh, *Frontend Intern* Dec 2019 - Feb 2020
Developed chatbot interfaces using AngularJS and Node.js with senior mentors.